

Dietary lignan intakes in relation to survival among women with breast cancer: the Western New York Exposures and Breast Cancer (WEB) Study

Dietary lignan intakes have been associated with reduced breast cancer risks; however, no previous studies have investigated whether lignan intake might be associated with breast cancer survival. We examined the association of dietary lignan intakes with survival in 1122 women with primary, incident, histologically confirmed breast cancer identified between 1996 and 2001, and with vital status determined through December 31, 2006. Diet in the 12–24 months before diagnosis was assessed with an extensive food frequency questionnaire, and potential confounders assessed from an extensive epidemiologic interview and abstracted clinical data. Lignan intake was calculated using published food composition data. Hazard ratios (HR), and 95% confidence intervals (CIs) for dietary lignan intakes with all cause, and breast cancer mortality were estimated using Cox proportional hazards adjusting for age, education, race, total energy intake, tumor stage, and body mass index. Of the 1122 women with complete dietary data, 160 had died by the end of follow-up. Among postmenopausal women only, those in the highest versus lowest quartile of lignan intakes had a statistically significant reduction in the risk of all cause mortality (HR 0.49, 95% CI 0.26–0.91) and a significantly reduced risk of breast cancer mortality (HR 0.29, 95% CI 0.11–0.76). Higher intakes of dried beans (HR 0.61, 95% CI 0.36–1.03), but not fruits, vegetables, or grains, were also weakly associated with overall mortality. In summary, our results suggest that higher lignan intakes may be associated with improved survival among postmenopausal women with breast cancer.